



Betsy Johanna Villa Brochero

Automation and Digital Transformation Engineer

Automation & Digital Transformation Engineer with experience driving Industry 4.0 initiatives in global manufacturing environments. Proven track record of improving factory performance through data-driven automation, AI-based inspection systems, and system integration. Strong background in operational excellence, cross-site collaboration, and technology roadmap development to enhance productivity, traceability, and cost efficiency.

✉ betsyvilla14@gmail.com

☎ +31-64-470-1628

📍 Enschede, Netherlands

🌐 [linkedin.com/in/betsy-villa-brochero](https://www.linkedin.com/in/betsy-villa-brochero)

WORK EXPERIENCE

Industrial Transformation Engineer (Global) Gunnebo Safe Storage

09/2023 - Present

Remote

The Gunnebo Group is a global provider of security products, services and software

- Drive digital transformation initiatives across multiple global manufacturing sites aligned with annual operational excellence targets.
- Develop and prioritize factory digitalization roadmaps including automation, data acquisition, and performance monitoring initiatives.
- Collaborate with Global Operational Excellence and plant leadership teams to identify improvement opportunities in productivity, quality, and traceability.
- Support local engineering teams in implementing scalable Industry 4.0 solutions aligned with business objectives.
- Build structured backlogs of automation and system-integration projects with defined implementation plans and KPIs.
- Facilitate cross-site knowledge sharing to standardize best practices in manufacturing digitalization.

Contact : David Aviram - David.Aviram@gunnebo.com

Automation and Digital Transformation Engineer Gunnebo Safe Storage

08/2022 - 08/2023

Netherlands

The Gunnebo Group is a global provider of security products, services and software

- Introduced Industry 4.0 and technological solutions including data acquisition and exchange, internet of things (IoT), business intelligence, and artificial intelligence.
- Created a road map to guide automation projects from conception to final implementation.
- Working within a team to find new improvements opportunities for the production process.

Contact : Renze Eringfeld - renze.eringfeld@gunnebo.com

SKILLS

Digital Transformation

Operational Excellence

Industrial Automation & Controls

Continuous Improvement

Machine Learning

Industry 4.0

Data-Driven Decision Making

TECHNICAL SKILLS

Programming

C++, Arduino, Matlab, Simulink, LabView, Java, Python, GIT, Github, Azure, OpenCV, ROS

Cloud Computing

Microsoft Azure Services

Virtual Factory Simulation

Simul8

EDUCATION

PhD in Computer Vision University Twente

09/2023 - Present

Enschede

- PhD Research focused on applied Computer Vision for real-time defect detection in industrial assembly processes, improving production reliability and quality assurance.

Master of Science in Mechatronics and Micromechatronics Systems Oviedo University/ Karlsruhe University of Applied Sciences

09/2018 - 09/2020

Spain/Germany

- Software, Robotics, Automotive, Mechatronics Systems, Artificial Intelligence, Control.
- Master Thesis: Control and Monitoring of an induction hardening test bench.

WORK EXPERIENCE

Trainee Digital Transformation Engineer Gunnebo Safe Storage

03/2021 - 07/2022

Netherlands

The Gunnebo Group is a global provider of security products, services and software

- Designed and implemented automated visual inspection systems for assembly processes.
- Developed machine learning algorithms for defect and missing-object detection in manufacturing lines.
- Created graphical user interfaces for operator interaction with inspection systems.
- Supported deployment and validation in real production environments.

Contact : Pim Bertou - pim.bertou@gunnebo.com

Mechatronics Intern Ikerlan S.Coop

02/2020 - 07/2020

Spain

R&D technology center specializing in product, process and service innovation helping companies enhance their competitiveness

- Developed LabVIEW-based control and monitoring system for induction hardening test bench.
- Selected and integrated sensors and actuators for control, power, and mechanical systems.
- Conducted technical feasibility and cost analyses for industrial implementation.

Contact : Unai Segurajauregi - usegurajauregi@ikerlan.es

PROJECTS

Visual Inspection Assembly System (VIAS)- Gunnebo (04/2021 - Present)

- Designed and deployed computer vision system to automate inspection in assembly processes.
- Improved inspection consistency and reduced manual verification workload.
- Enabled data-driven quality analysis for continuous improvement initiatives.
- Integrated solution into production environment for real-time monitoring.
- Utilized Machine Learning to enhance the inspection process at the factory.

Synaps (Digital Environment Production)- Gunnebo (04/2021 - 09/2023)

- Led integration of manufacturing systems into centralized digital platform.
- Enabled real-time production data analysis and improved cross departmental visibility.
- Enhanced traceability and resource utilization through system consolidation.

REFERENCES

Gilbert Koenders, Manager Production Development,
Global Operational Excellence Manager

Contact : gilbert.koenders@gunnebo.com

EDUCATION

Bachelor of Science in Mechatronics Engineering

Autonoma del Caribe, University

01/2010 - 03/2015

Colombia

- Control, Electronics, Digital Image Processing, Intelligent Systems, Industrial Automation.
- Bachelor Thesis : Digital image processing applied on static sign language recognition system.

LANGUAGES

English

Native or Bilingual Proficiency

Dutch

Limited Working Proficiency

German

Limited Working Proficiency

French

Elementary Proficiency

Spanish

Native or Bilingual Proficiency

HONOR & AWARDS

Fellowship: Multidisciplinary Artificial Intelligence Training Program (08/2021 - 04/2022)

IALAB- Buenos Aires University, Argentina

- Programming in Artificial Intelligence Online Training, University of Buenos Aires. (08/2021-03/2022). Funded by the Development Bank of Latin America and the University of Buenos Aires, Argentina.

Fellowship : Joint Master Erasmus Mundus in Mechatronics (09/2018 - 09/2020)

European Union

- Joint Master in Mechatronics Engineering, Erasmus Mundus Program. Scholarship funded by the European Union to Study a Master in Mechatronics at Karlsruhe University of Applied Sciences (Germany) and University of Oviedo (Spain)

DAAD Scholarship for study trips to Germany (09/2014 - 10/2014)

DAAD

- Scholarship for an academic mission and visit different research centers and universities in Germany.

CERTIFICATES

AI for Everyone (02/2023 - 04/2023)

Deep Learning Academy

AI In Manufacturing (09/2022 - 10/2022)

University of Delft - Netherlands